

JUNMING LIU

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EDUCATION

Tongji University, Shanghai, China

Setp. 2023 – Mar. 2026

M.E. in Computer Science and Technology

- GPA: 90.0/100
- Research Interests: Generative Intelligence, Multimodal Reasoning, and Graph Theory

Dalian Maritime University, Dalian, China

Sept. 2019 – June 2023

B.E. in Intelligence Science and Technology

- GPA: 84.0/100

PUBLICATIONS

* EQUAL CONTRIBUTION † CORRESPONDING AUTHOR

- [1] **Junming Liu**, Siyuan Meng, Yanting Gao, et al. “Aligning Vision to Language: Annotation-Free Multimodal Knowledge Graph Construction for Enhanced LLM Reasoning.” *ICCV*, 2025 (Accepted).
- [2] **Junming Liu**, Yanting Gao, Yuqi Li, et al. “Mosaic: Data-Free Knowledge Distillation via Mixture-of-Experts for Heterogeneous Distributed Environments.” *Knowledge-Based Systems*, 2026 (Accepted).
- [3] **Junming Liu**, Yifei Sun, Weihuang Cheng, et al. “ReBrain: Brain MRI Reconstruction from Sparse CT Slice via Retrieval-Augmented Diffusion.” *WACV*, 2026 (Accepted).
- [4] **Junming Liu**, Yusen Zhang, Rongchao Zhang, et al. “Domain-Adaptive Model Merging Across Disconnected Modes.” *ICASSP*, 2026 (Accepted).
- [5] Yujin Kang*, **Junming Liu***, Haiyan Cui†. “AI-Driven Assessment of Lip Volume Improvement Using Hyaluronic Acid Fillers: A Comprehensive Analysis.” *Aesthetic Plastic Surgery*, 2025 (Accepted).
- [6] Aoqi Wu, **Junming Liu**, Yuwei Zhang, et al. “AMID: Model-Agnostic Dataset Distillation by Adversarial Mutual Information Minimization.” *WWW*, 2026 (Accepted).
- [7] Yanting Gao, Yepeng Liu, **Junming Liu**, et al. “Boosting Adversarial Transferability via Commonality-Oriented Gradient Optimization.” *PRCV*, 2025 (Accepted).
- [8] Pei Liu, Xin Liu, Ruoyu Yao, **Junming Liu**, et al. “HM-RAG: Hierarchical Multi-Agent Multimodal Retrieval Augmented Generation.” *ACM MM*, 2025 (Accepted).
- [9] Yuqi Li*, Siwei Meng*, Chuanguang Yang, Weilun Feng, **Junming Liu**, et al. “A Comprehensive Survey of Interaction Techniques in 3D Scene Generation.” *IJCAI*, 2026 (Accepted).
- [10] Yuqi Li*, Junhao Dong*, Jiao Liu, Piotr Koniusz, Hansheng Zeng, Chuanguang Yang, **Junming Liu**, et al. “Evolving Multimodal Models for Physical Dynamics: A Multi-objective Neuroevolution Approach.” *IEEE Transactions on Evolutionary Computation*, 2026 (Accepted).
- [11] **Junming Liu**, Yuqi Li, Yifei Sun, et al. “Self-Evolving Spatial Reasoning in Vision Language Models via Geometric Logic Consistency.” *NeurIPS*, 2026 (Under Review).
- [12] Rongchao Zhang, **Junming Liu**, Haoran Liu, et al. “Be an Artist Who Revises: Learning Fluorescence Microscopy Characterization with Non-Instantaneous Flow Matching.” *NeurIPS*, 2026 (Under Review).
- [13] **Junming Liu***, Yifei Sun*, Weihua Cheng, et al. “MemVerse: Multimodal Memory for Lifelong Learning Agents.” *EMNLP*, 2026 (Under Review).
- [14] **Junming Liu***, Yuqi Li*, Shiping Wen, et al. “Hit-RAG: Learning to Reason with Long Contexts via Preference Alignment.” *IEEE Transactions on Artificial Intelligence* (Under Review).
- [15] **Junming Liu**, Yuqi Li, Yanting Gao, et al. “FedRecon: Missing Modality Reconstruction in Heterogeneous Distributed Environments.” *IEEE Transactions on Multimedia* (Under Review).
- [16] **Junming Liu**, Yifei Sun, Weihua Cheng, et al. “Hierarchical Memory Orchestration for Personalized Persistent Agents.” *IUI*, 2027 (Under Review).
- [17] Siyuan Meng*, **Junming Liu***, Yirong Chen, et al. “MemSelector: Plug-and-Play Dynamic Evidence Selection for Memory-Grounded Reasoning.” *WSDM*, 2026 (Under Review).

HONORS AND AWARDS

- **National College Mathematics Competition, National First Prize (1%)** *Dec. 2022*
Chinese Mathematical Society
- **Excellent Student Scholarship (10%)** *Sept. 2021, 2022*
Dalian Maritime University
- **Leadership and Communication Scholarship** *Sept. 2020*
Dalian Maritime University

RESEARCH EXPERIENCE

- **Reasoning in Multimodal Large Language Models** **Shanghai AI Laboratory**
Advisor: Dr. Ding Wang *Jan. 2025 – Present*
 - **Overview:** Worked on improving reasoning, memory, and world modeling capabilities of multimodal foundation models.
 - **Research:** Explored multimodal reasoning systems across five directions: (i) medical foundation model mid-training for domain adaptation, (ii) post-training strategies for spatially-aware VLM reasoning, (iii) JEPA-based world modeling for predictive representation learning, (iv) retrieval-augmented generation for knowledge-intensive reasoning, and (v) memory-augmented agents for long-horizon interaction.
 - **Outcome:** Improved robustness and generalization across multiple multimodal benchmarks, with several works accepted in top-tier conferences and others under review.
- **Multimodal Generative Modeling for Cellular Perturbation** **Stanford University**
Advisor: Dr. Dongxia Wu *Apr. 2026 – Present*
 - **Overview:** Studied multimodal generative modeling for joint modeling of cell morphology and gene expression under drug perturbations.
 - **Research:** Developed a flow-matching framework with cross-modal fusion to model non-linear dynamics between imaging and expression without information loss.
 - **Outcome:** Achieved high-fidelity mapping of phenotypic responses, establishing a robust generative baseline for computational pharmacology.
- **Vision-Language Models for Medical Imaging** **Tongji University**
Advisor: Prof. Guosun Zeng *Sept. 2023 – Mar. 2026*
 - **Motivation:** Brain medical images contain complex structures that are difficult to interpret reliably, motivating the use of VLMs to enable structured and interpretable reasoning.
 - **Methods:** Used DDPM with ControlNet to perform high-fidelity reconstruction that preserves anatomical consistency across modalities, fine-tuned VLMs for brain abnormality detection and segmentation with traceable and interpretable reasoning, and incorporated privacy-preserving, distributed techniques for efficient lightweight collaboration of large models across heterogeneous hospitals.
 - **Results:** Achieved improved modality-aligned reconstruction and enhanced brain tumor detection and segmentation on BraTS, etc., demonstrating strong robustness even under Non-IID data distributions.
- **Text Sentiment Analysis Based on BERT Model** **Dalian Maritime University**
Advisor: Prof. Yijia Zhang *Sept. 2021 – May. 2022*
 - **Motivation:** Investigate public opinion tendencies on social media during the COVID-19 pandemic.
 - **Methods:** Enhanced BERT by integrating LDA topic modeling and fine-tuned the combined model for text sentiment analysis.
 - **Results:** Achieved 95% sentiment classification accuracy on social media datasets, enabling robust tracking of public emotional trends.

INTERNSHIP EXPERIENCE

- **Shanghai AI Laboratory** *Jan. 2025 – Present*
 - Conducted research on AI agents.
- **Beijing Jinxixin Network Technology Co., Ltd.** *Oct. 2024 – Dec. 2024*
 - Constructed and trained vertical domain large models for maritime law scenarios.

- **Shanghai NIO Automobile Co., Ltd.**

June. 2023 – Aug. 2023

- Controlled steering systems using multimodal sensing for rollover prevention in tire blowout scenarios.

SKILLS

- Language: Chinese (native), English (IELTS: 7.0), Japanese, German
- Programming: C, C++, Python, Java, Go, SQL, Rust, MPI, NCCL, DeepSpeed, DDP, FSDP

CAMPUS EXPERIENCE

- Member of the Student Union, School of Electronics and Information Engineering, Tongji University
Sept. 2023 – June. 2024
- Member of the Art Troupe, School of Information Science and Technology, Dalian Maritime University
Sept. 2019 – June. 2022